COUT AND ESCAPE SEQUENCE

1.

```
cout<<"welcome to a \network program ";
cout<<"written by \tamer \r ali \nora\bsarah ";
cout<<"endl\n";
cout<<endl;
cout<<"welcome \to \"C++\'\\";
cout<<"/n";
```

2.



3.

```
int x=20,y=2;
cout<<“C++”;
cout<<x;
cout<<y;
```

```
C++202
```

4.

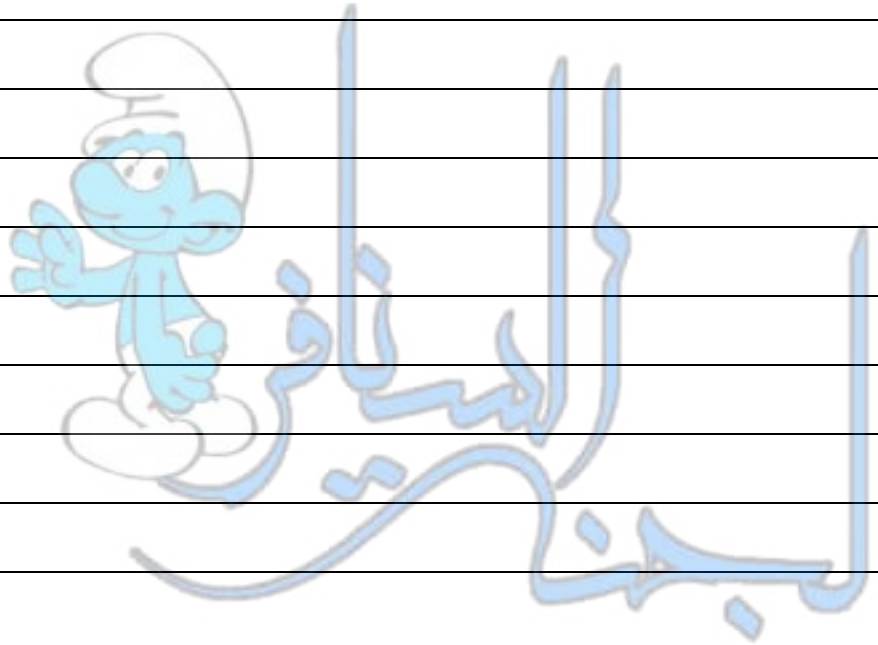
```
int x=20,y=2;
cout<<“C++”<<endl;
cout<<x<<endl;
cout<<y<<endl;
```

```
C++
20
2
```

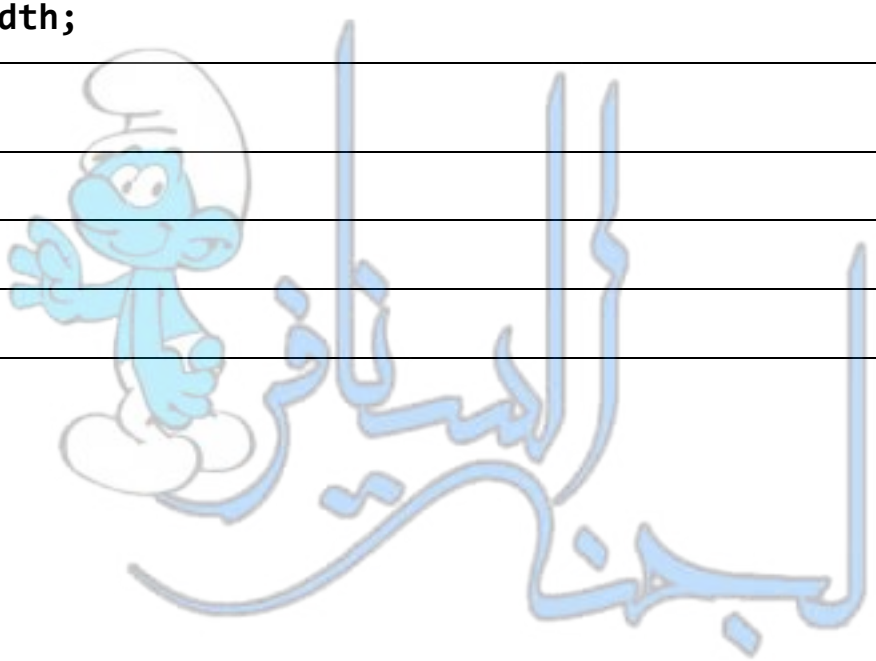
5.	**int x=20,y=2;** **cout<<"C++"<<endl<< x<<endl <<y<<endl;**	**C++** **20** **2**
6.	**int x=20,y=2;** **cout<<"the sum"<<x+y<<endl;** **cout<<y<<endl;**	**the sum 22** **2**



IDENTIFIERS		
WHICH OF THE FOLLOWING IS A VALID C++ IDENTIFIER		
1.	const	X
2.	y=z	X
3.	Ali	√
4.	ahmad’s	x
5.	two-way	x
6.	int	x
7.	my sum	x
8.	my_sum	√
9.	_average	√
10.	2ndclass	x



ASSIGNMENT STATEMENT		
WHICH OF THE FOLLOWING IS A VALID ASSIGNMENT STATEMENT		
1.	`length=20;`	**√**
2.	`width=30;`	**√**
3.	`area=length*width;`	**√**
4.	`r=3;` `y=r*2+3;`	**√**
5.	`20=length;`	**x**
6.	`length=20`	**x**
7.	`x+y=8;`	**x**



ARITHMETIC OPERATORS		

1.	int m=10;
	int s=3;
	int r;
	r=m+s;
	r=m-s;
	r=m*s;
	r=m/s;
	r=m%s;
		-
		-
		-
		r=13
		r=7
		r=30
		r=3
		r=1

2.	int m=10;
	int s=3;
	float r;
	r=m+s;
	r=m-s;
	r=m*s;
	r=m/s;
	r=m%s;
		-
		-
		-
		r=13.0
		r=7.0
		r=30.0
		r=3.0
		r=1.0

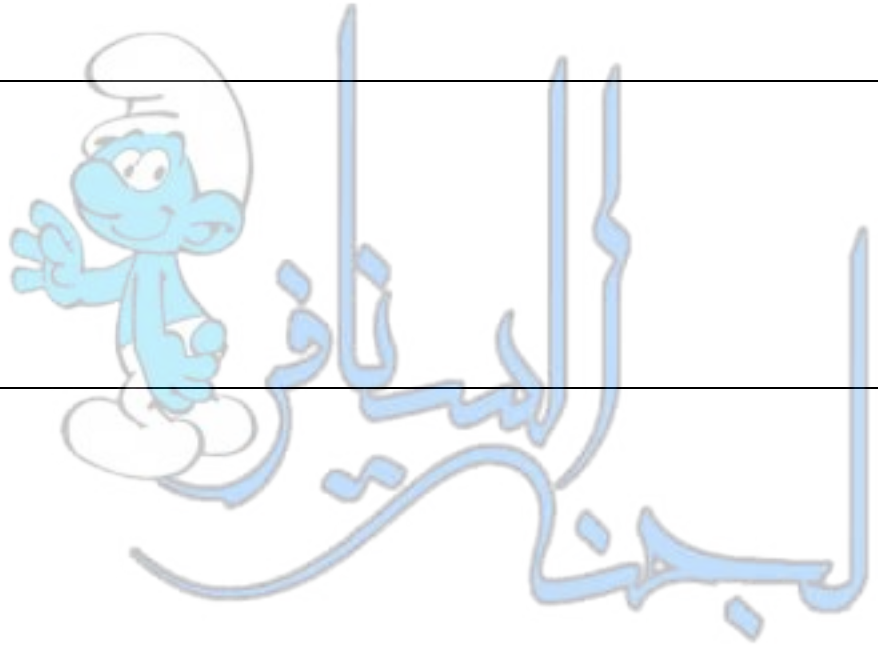
3.	float m=10;
	float s=3;
	int r;
	r=m+s;
	r=m-s;
	r=m*s;
	r=m/s;
		error

	r=m%s;	

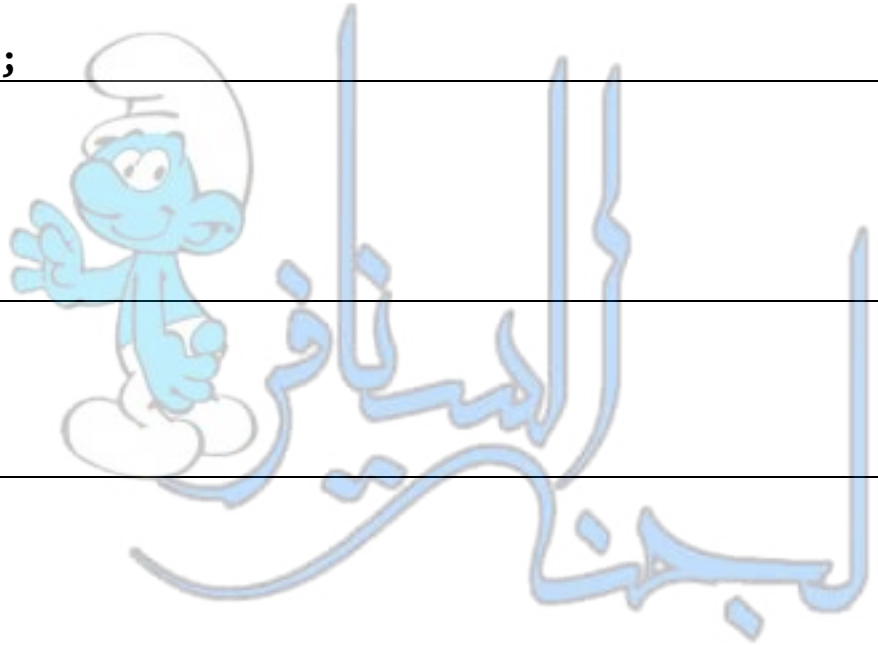
4.	float m=10;
	float s=3;
	int r;
	r=m+s;
	r=m-s;
	r=m*s;
	r=m/s;
		-
		-
		-
		r=13
		r=7
		r=30
		r=3

5.	float m=10;
	float s=3;
	float r;
	r=m+s;
	r=m-s;
	r=m*s;
	r=m/s;
		-
		-
		-
		r=13.0
		r=7.0
		r=30.0
		r=3.33333

6.	int m=10;
	int s=3;
	float r;
	r=m+s;
	r=m-s;
	r=m*s;
	r=float (m)/s;
	r=m%s;
		-
		-
		-
		r=13.0
		r=7.0
		r=30.0
		r=3.33333
		r=1.0



7.	**char c='A';** **c=c+1;** **cout<<c<<endl;** **cout<<c+1;** **char b=c+3;** **cout<<b;**	**B** **67E**
8.	**int x='a';** **cout<<x<<endl;** **cout<<char (x);**	**97** **a**
9.	**bool b;** **b=50;** **cout<<b<<endl;** **b=false;** **cout<<b;**	**1** **0**
10.	**float x=5;** **int y=2;** **int z=x/2;** **cout<<z;**	**2**



PRIORITY RULES		

1.	int a=2,b=2,c=3,r;
	r=b*2-4*a*c;
	cout<<r;
		-20

2.	int a=2,b=2,c=3,r;
	r=b+a+c*b*a/6;
	cout<<r;
		6

3.	int a=2,b=2,c=3,r;
	r=c%a+b*c/a;
	cout<<r;
		4

4.	int a=2,b=3,c=4,r;
	r=a*(9+(b+5)/2)-c;
	cout<<r;
		22

5.	int a=2,b=3,c=4,r; r=17%(c-a)*b+6 ; cout<<r;	9

6.	int a=3,b=8,r;
	r=(a>=0)+(a!=b);
	cout<<r;
		2

7.	**int a=3,b=8,r;** **r=(a+5*b)<=(30/a);** **cout<<r;**	**0**
8.	**int a=3,b=8,r;** **r=a+b+3<b+a !=30;** **cout<<r;**	**1**



COMBINED OPERATORS		

1.	int x=9;
	int y=6+ --x;
	cout<<x<<endl;
	cout<<y;
		8
		14

2.	int x=9; int y=6+ x--; cout<<x<<endl; cout<<y;	8 15

3.	int x=3,z=5,k=10;
	k-=++x - z-- + k++;
	cout<<x<<endl<<k<<endl<<z;
		4
		2
		4

4.	int x=5,y=0,z=3;
	z+= ++x - y++ + z-- +x++;
	cout<<x<<endl<<y<<endl<<z;
		7
		1
		17

5.	int x=5,y=0,z=3;
	z+= ++x + y++ -z++;
	cout<<z<<endl<<x<<endl<<y;
		7
		6
		1

RELATIONAL AND LOGICAL OPERATORS		

1.	int i=10,j=15,r;
	r=i>j;
	cout<<r;
		0

2.	int i=10,j=15,r;
	r=!(i>j);
	cout<<r;
		1

3.	int i=10,j=15,r;
	r=(i>=j) && (i==j);
	cout<<r;
		0

4.	int i=10,j=15,r;
	r=(i>=j) && (i<j);
	cout<<r;
		0

5.	int i=10,j=15,r;
	r=(i<=j) && (i!=j);
	cout<<r;
		1

6.	int i=10,j=15,r;
	r=(i>=j) || (i==j);
	cout<<r;
		0

7.	int i=10,j=15,r;
	r=(i>=j) || (i<j);
	cout<<r;
		1

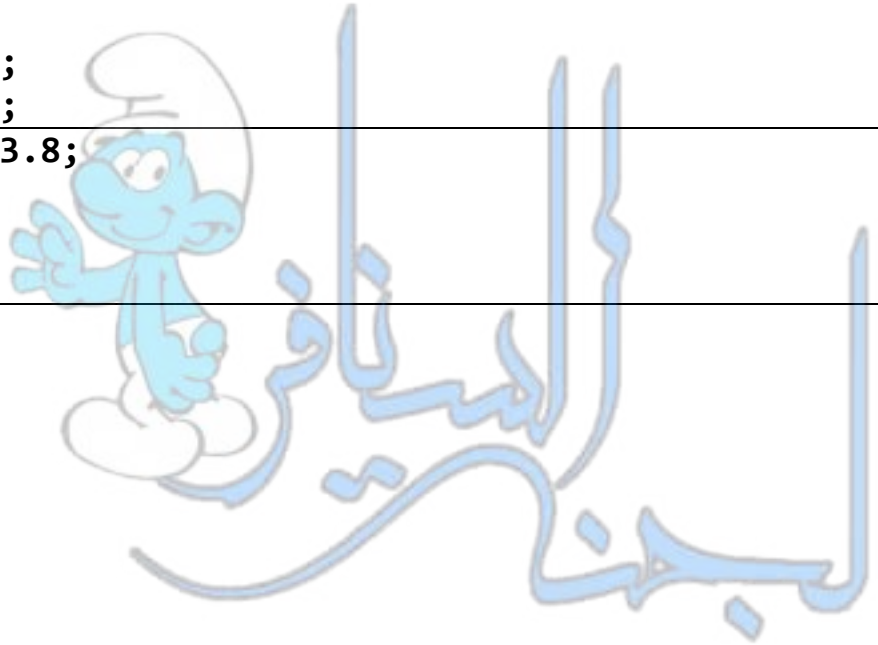
8.	int i=10,j=15,r;	1

	r=(i<=j) || (i!=j); cout<<r;	
9.	int x=-2; cout<<!(x||1);	0

10.	int x=10,y=20,z=5,i=5;
	i+=(x<y && y<z) || (x!=z);
	cout<<i;
		6

11.	bool x; x=(true && !2); cout<<x<<!x+24;	025

12.	const float w=3.8;
	float d=1.8;
	w-=d;
	cout<<w;
		error



IF STATEMENT		

1.	int x=5,y=10,z=4;
	if (x) z=y/x;
	cout<<x<<z;
		52

2.	int x=5,y=10,z=4;
	if (!x) z=y/x;
	cout<<x<<z;
		54

3.	int x=5,y=10,z=4;
	if (x==5) z=y/x;
	cout<<x<<z;
		52

4.	int x=5,y=10,z=4;
	if (x=6) z=y/x;
	cout<<x<<z;
		01

5.	int x=5,y=10,z=4;
	if (x=0) z=y/x;
	cout<<x<<z;
		04

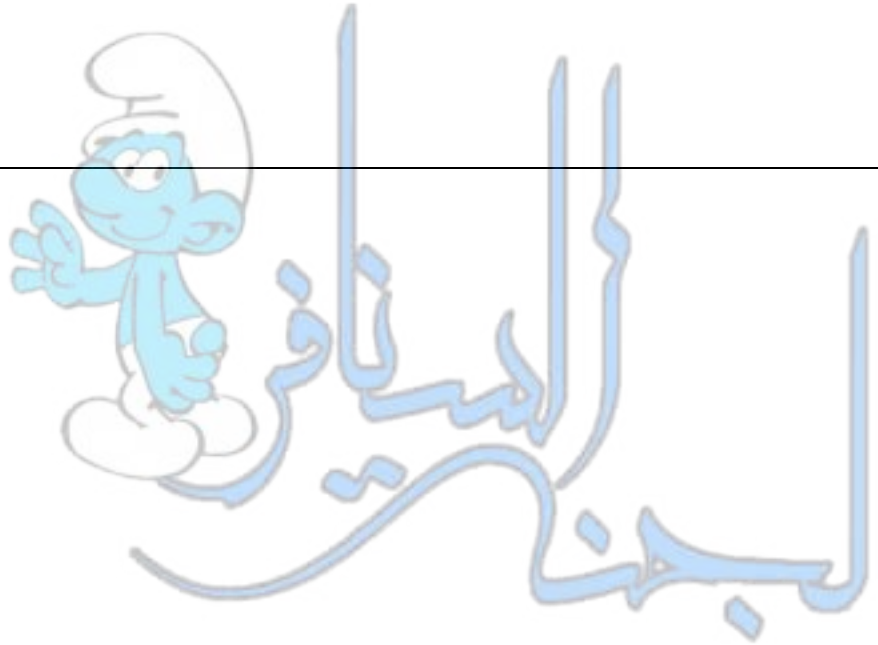


IF ELSE		
1.	int x=3; if (x<=3) if (x>=3) cout<<"x"; else cout<<"y"; cout<<"z";	xz
2.	int m=5,n=3; if (m>n) cout<<'A'; else cout<<'B'; cout<<'C';	AC

3.	int one=3,two=5;
	if(one < two && 1)
	cout<<one;
	else if (!one && (two %2))
	{
	cout <<++one ;
	cout<<--one;
	}
	if (5 % !(!one))
	cout<<2;
	else
		32

	if(0) cout<<1; else cout<<2;	

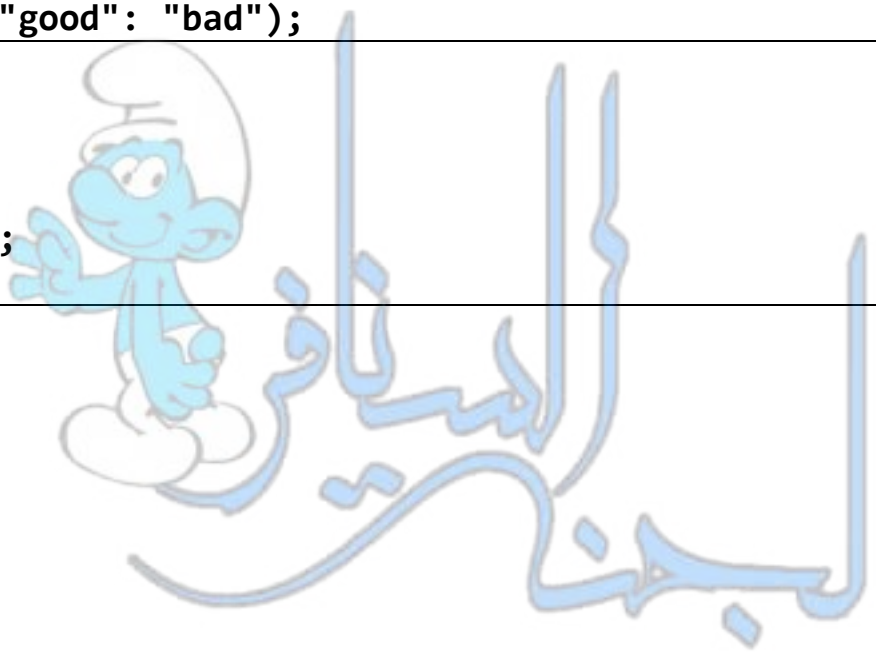
4.	int a=-3,b=4,c=0;
	if(a>0)
	if(b>0)
	c=2;
	else
	c=3;
	cout<<c;
		0



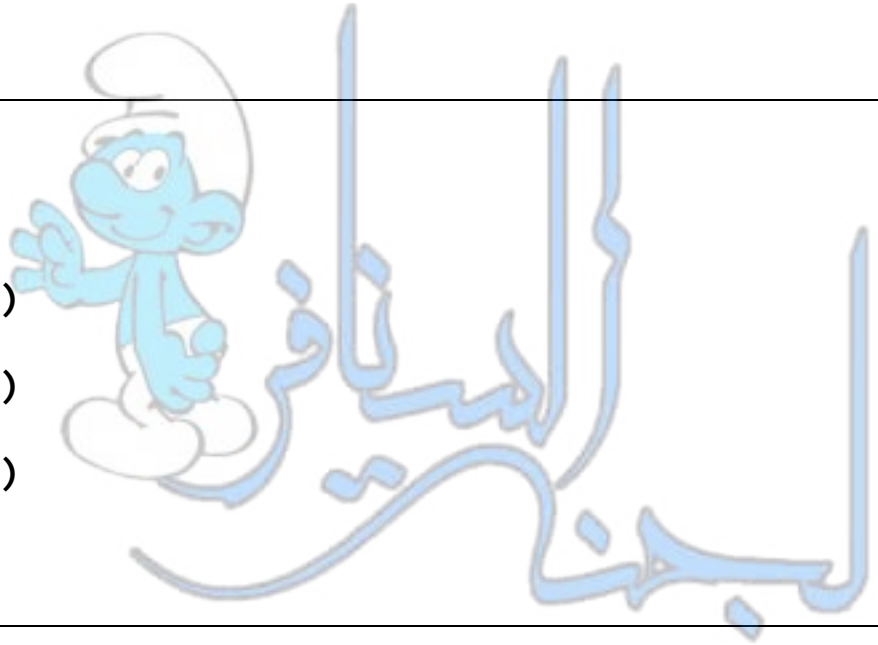
CONDITIONAL OPERATOR		

1.	int a ,b, c,r;
	a=5;
	b=10;
	c=7;
	cout<<(a==b ? "good": "bad");
		bad

2.	int a ,b, c,r;
	a=5;
	b=10;
	c=7;
	r=a<b ? c: a+b;
	cout<<r;
		7



NESTED IF ELSE		
1.	int x=2,y=3,z=0; if(x>y || !x) cout<<x; else if (z) cout<<y*x; else cout<<y;	3
2.	int g; cin>>g; if (g>=90) cout<<'A'; else if (g>=80) cout<<'B'; else if (g>=70) cout<<'C'; else if (g>=50) cout<<'D'; else cout<<'F';	



SWITCH EXAMPLES

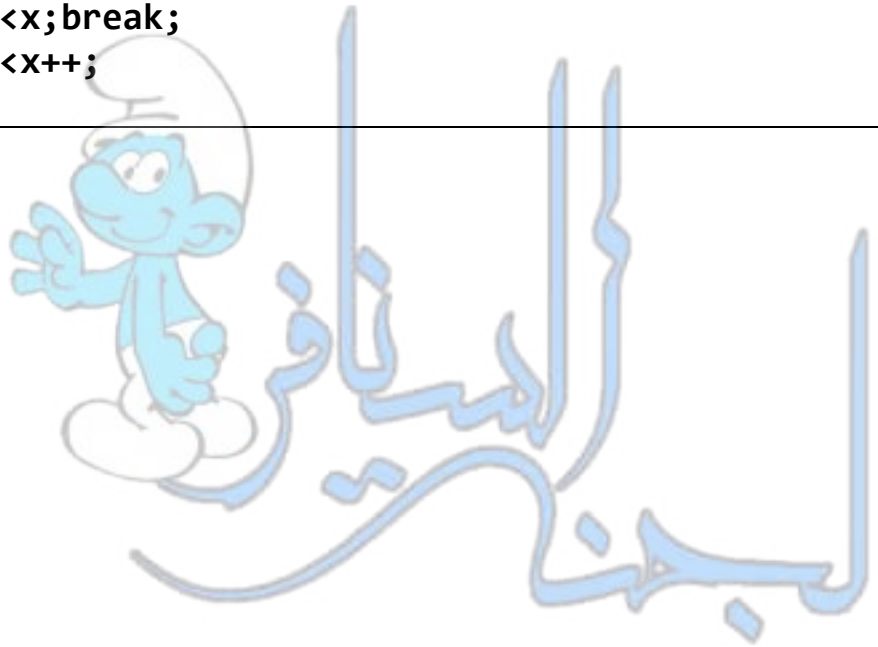
1.	`int num;`
	`cin>>num;//1,3,5`
	`switch(num)`
	`{case 1:`
	`case 2:cout<<"x";break;`
	`case 5:cout<<"y";`
	`default:cout<<"z";}`
	`cout<<"*";`
		`num=1 → x*`
		`num=3 → z*`
		`num=5 → yz*`

2.	`char x;`
	`cin>>x;`
	`switch(x)`
	`{case 'a':cout<<'A';`
	`case 'b':cout<<'B';`
	`case 'c':cout<<'C';`
	`case 'd':cout<<'D';}`
	`cout<<"end";`	

3.	`int i,j;`
	`cin>>i;//i=9,i=3`
	`switch(i/4)`
	`{case 0: j=-i;`
	`case 1:j=2*i;break;`
	`case 2:j=i+5;`
		`i=9 →j=-4`
		`i=3 →j=6`

	default:j+=-2*i;} cout<<“j=“<<j;	

4.	char x;
	cin>>x;//x=’b’,’e’
	switch(--x+2)
	{default:cout<<--x;
	case ‘c’:cout<<x+3;
	case 100:cout<<x;break;
	case ‘B’:cout<<x++;
	}
		x=’b’ → 100a
		x=’e’ →
		c102c



WHILE EXAMPLES

1.

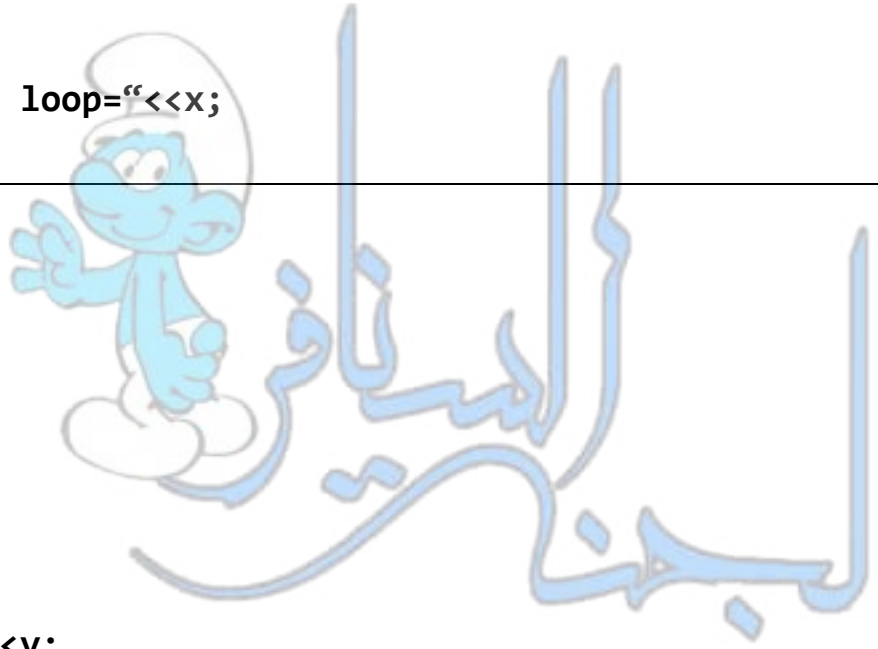
```
int x=20;
while (x!=0)
{cout<<"x="<<x<<endl;
x-=5;
}
cout<<"x after loop="<<x;
```

```
x=20
x=15
x=10
x=5
x after loop=0
```

2.

```
int y=4;
int z=4;
while(z<5)
{int a=8;
z+=a++;
z++;
cout<<a<<endl;
break;
}
y=4*z;
cout<<z<<endl<<y;
```

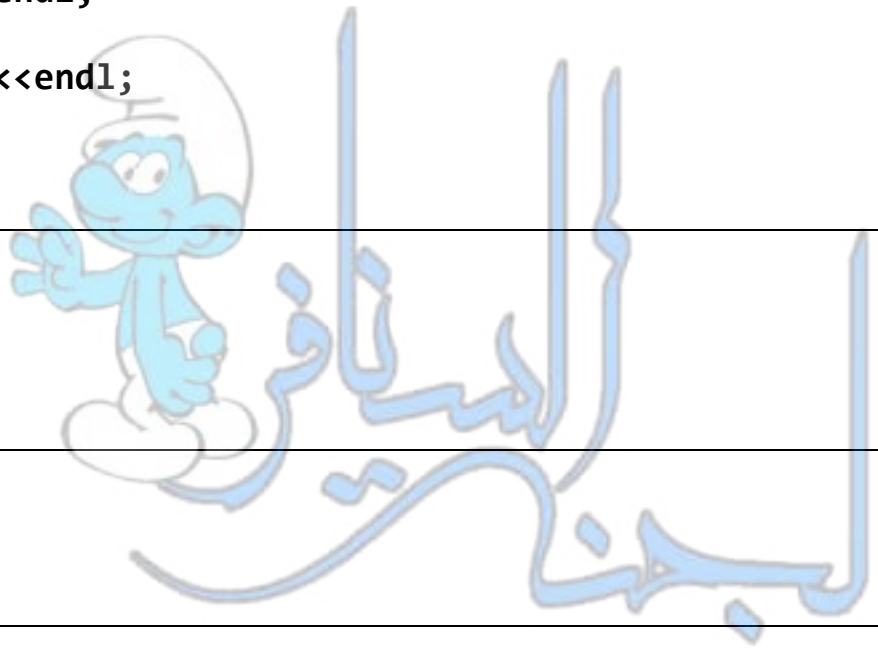
```
9
13
52
```



3.	int x=0;
	while(x<3)
	{ switch (x+2)
	{
	case 3:
	cout<<x<<endl;
	case 0:
	cout<<x-1<<endl;
	break;
	}
	x++;}
		1
		0

4.	int n=1;
	while (n<=5)
	cout<<n;
	n++;
		1111111……
		infinite loop

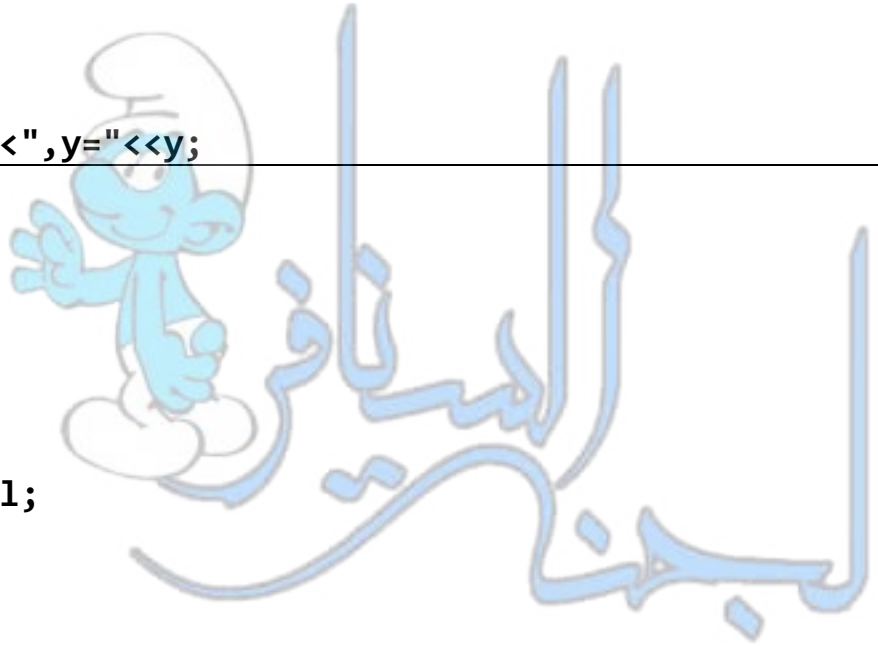
5.	while(-19); cout<<"hi";	No output keeps doing nothing for infinity



DO WHILE EXAMPLES

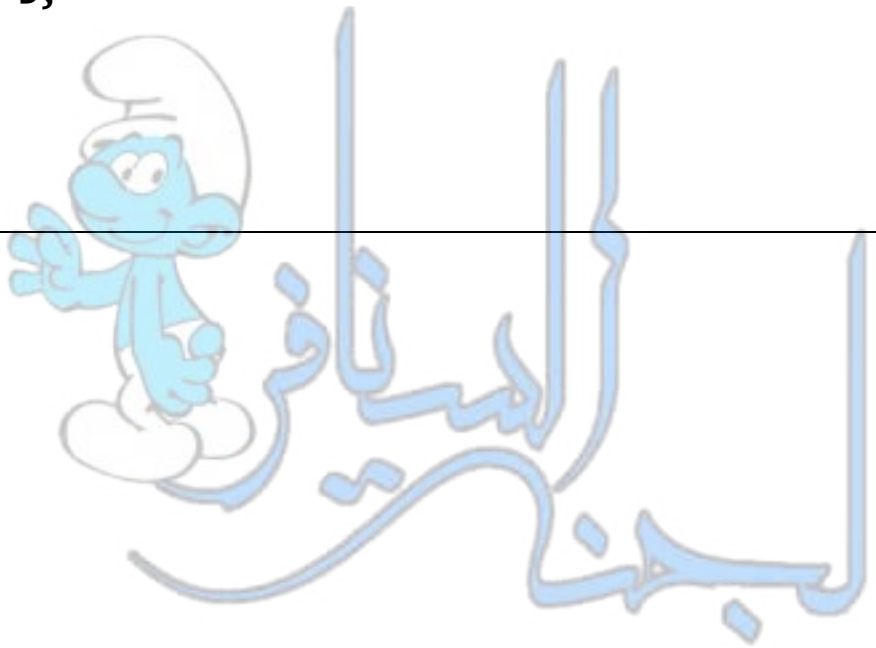
1.	`int x=1,y=5;`
	`do{`
	`if(y%x!=0)`
	`cout<<x++ +y;`
	`x++;`
	`y--;`
	`}while(y>=x);`
	`cout<<"x="<<x<<",y="<<y;`
		`x=4,y=2`

2.	`int a=3;`
	`char g='f';`
	`do{`
	`a++;`
	`if(g=='m')`
	`cout<<g<<endl;`
	`else`
	`cout<<++a<<endl;`
	`g='m';`
	`}while(a<8);`
	`cout<<"***";`
		`5`
		`m`
		`m`
		`m`
		`***`



3.	int i=3;
	do{
	cout<<i;
	i--;
	}while(i<0);
		3

4.	int r=10,c=2,n=3;
	do{
	r*=2;
	c++;
	}while(c<=n);
	cout<<r;
		40



FOR EXAMPLES

1.	for(int j=2;j<=20;j+=3) cout<<j<<"\t";	2 5 8 11 14 17 20
2.	for(int i=0,j=0;i+j<=10;i++,j++) cout<<i+j<<endl;	0 2 4 6 8 10

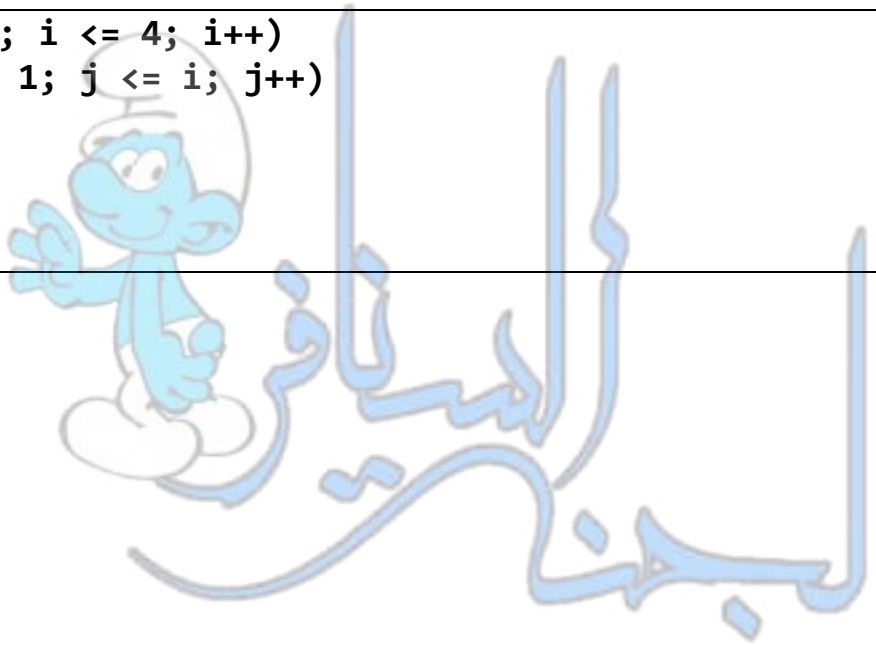
3.	for(int a=0;a<=10;a+=2)
	{
	int c=a%3;
	switch(c)
	{case 0: a=a+1;break;
	case 1: a=a+4;break;
	default: a=a/2;}
	cout<<"a="<<a<<endl;}
		a=1
		a=4
		a=7
		a=10

4.	for(int x=2;x<4;x--) {if(x<2) break;	-4

	else if(x==2) x=-4; cout<<x;}	
5.	for(int x=0;x<=20;x+=2) if(x%2==0 && x%5==0) cout<<"x="<<x<<endl;	x=0 x=10 x=20

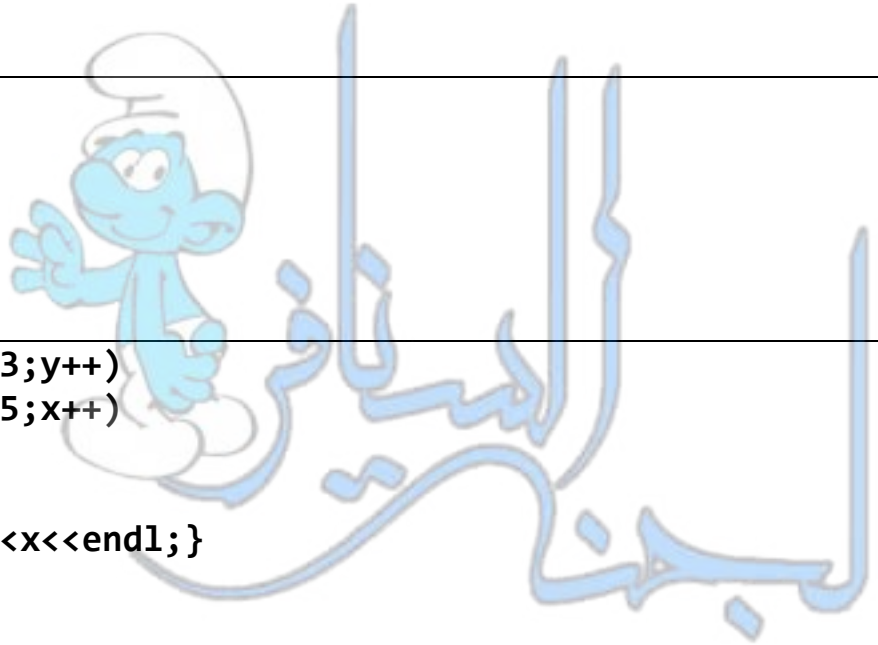


NESTED FOR EXAMPLES		
1.	for(int j=3;j<=5;j++) {for(int i=1;i<=2;i++) cout<<i*j<<'\t'; cout<<endl;}	3 6 4 8 5 10
2.	for (int i = 1; i <= 4; i++) { for (int j = 1; j <= i; j++) cout<<"*"; cout<<endl; }	* ** *** ****



BREAK & CONTINUE EXAMPLES		
1.	int x=1; while(x<=5) {++x; if(x==3) break; cout<<x;}	2
2.	int x=1; while(x<=5) {++x; if(x==3) continue; cout<<x;}	2456

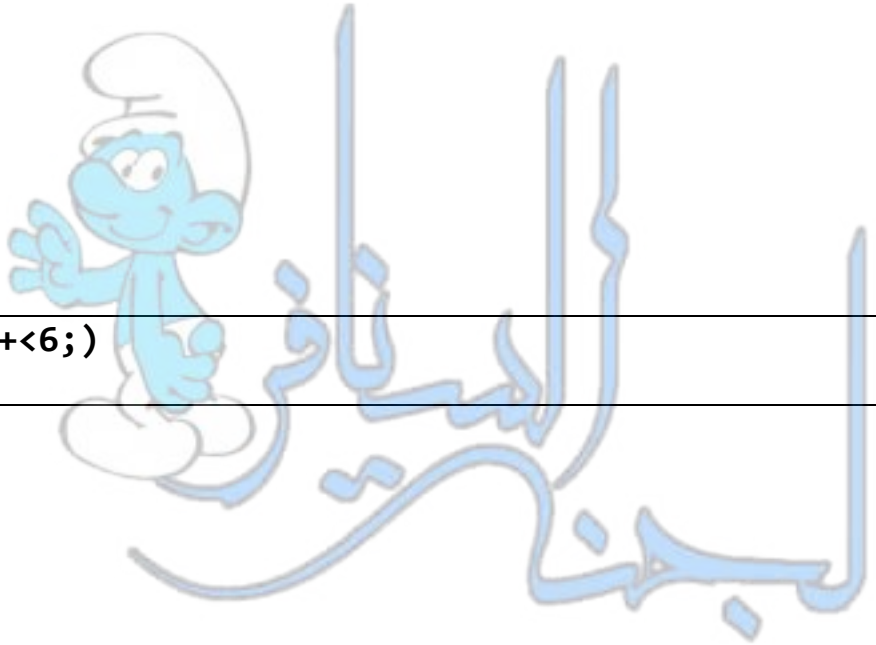
3.	for(int y=0;y<3;y++)
	for(int x=0;x<5;x++)
	{
	if(x>=2)break;
	cout<<y<<'\t'<<x<<endl;}
		0 0
		0 1
		1 0
		1 1
		2 0
		2 1



4.	**for(int x=1;x<=5;x++) {if(x==3) continue; cout<<x;}**	**1245**

5.	**int i=0;
	do{
	i++;
	cout<<"+++\n";
	cout<<"***\n";
	continue;
	cout<<"C++";
	}while(i<2);
	cout<<"end";**
		**+++
		***
		+++
		***
		end**

6.	**for(int i=3;i++<6;) cout<<i;**	**456**



ONE DIMENSIONAL ARRAY		

1.	int a=3,b=2;
	int z[6]={1,2,3,4,5,6};
	z[a+b]+=2;
	z[0]=2;
	z[4]=(a/b+6);
	for(int i=0;i<6;i++)
	cout<<z[i];
		223478

2.	char x[5]={'a','h','m','a','d'}; for(int i=0;i<=4;i++) {x[i]++; cout<<x[i];}	binbe
3.	int v[5]={7,8,3,4}; cout<<v[3]<<endl<<(v[0]*v[2]++)<<endl<<++v[1]; cout<<endl<<v[2];	4 21 9 4

4.	`int z[]={5,7,9,3,4,8,1};`
	`for(int k=0;z[k]%3 !=0;k++)`
	`cout<<k<<z[k]/2<<endl;`
		02
		13

5.	`int a[4]={8,2,5,3};`
	`for(int i=1;i<=3;i++)`
	`{ for(int j=1;j<=3;j++)`
	`cout<<++a[j]<<"\t";`
	`cout<<endl; }`
		3 6 4
		4 7 5
		5 8 6

6.	`int L[10]={2,1,3,5,2,3,1,4,8,7};`
	`cout<<L[2]<<endl;`
	`cout<<L[L[2]]<<endl;`
	`cout<<L[L[2]+L[3]++]<<endl;`
	`cout<<L[L[L[2]]]<<endl;`
	`cout<<L[3]<<endl;`
		3
		5
		8
		1
		6

TWO DIMENSIONAL ARRAY

1.

```
int a[3][3]={{3,2,3},{4,4,2},{7}};
     for(int i=0;i<3;i++)
     { for(int j=0;j<3;j++)
     cout<<a[i][j]<<"\t";
     cout<<endl;   }
     int sum1=0,sum2=0;
     for (int i=0;i<3;i++)
          for ( int j=0;j<3;j++)
               if (i==j)   sum1+=a[i][j];
               else  if (i+j==2)
sum2+=a[i][j];
     cout<<"sum 1="<<sum1<<endl;
     cout<<"sum 2="<<sum2<<endl;
```

```
3   2   3
4   4   2
7   0   0
Sum1=7
Sum2=10
```

2.

```
#include <iostream>
using namespace std;
int main()
{int test[3][2] ={{2, -5},{4, 0},{9, 1}};
for(int i = 0; i < 3; ++i)
{for(int j = 0; j < 2; ++j)
{cout<< "test["<<i<<"]["<<j<<"] ="<<test[i][j]<<endl;        }
}
return 0;}
```

```
test[0][0] = 2
test[0][1] = -5
test[1][0] = 4
test[1][1] = 0
test[2][0] = 9
test[2][1] = 1
```

3.	#include <iostream>
	using namespace std;
	
	int main () {
	int a[5][2] = { {0,0}, {1,2}, {2,4}, {3,6},{4,8}};
	
	for (int i = 0; i < 5; i++)
	for (int j = 0; j < 2; j++) {
	
	cout << "a[" << i << "][" << j << "]: ";
	cout << a[i][j]<< endl;
	}
	
	return 0;
	}
		a[0][0]: 0
		a[0][1]: 0
		a[1][0]: 1
		a[1][1]: 2
		a[2][0]: 2
		a[2][1]: 4
		a[3][0]: 3
		a[3][1]: 6
		a[4][0]: 4
		a[4][1]: 8

STRINGS		
1.	char a[5]="ahmad";	error
2.	char a[]="ahmad";	
3.	char a[10]={'a','b','c','\0','d','e'}; cout<<a;	abc
4.	char a[10]={'a','b','c','\0','d','e'}; for(i=0;i<10;i++) cout<<a[i];	abc de

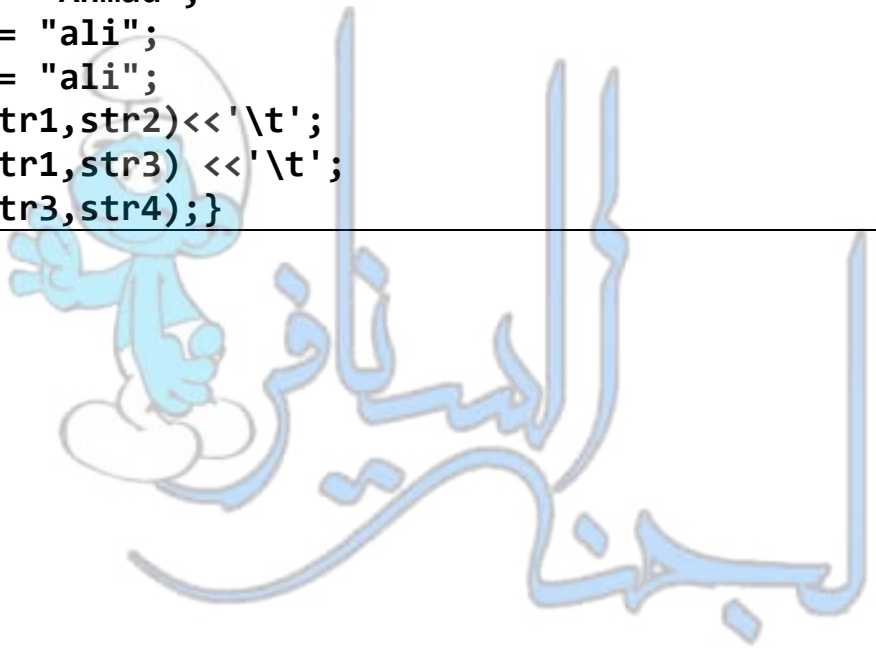
5.	cin>>a;//ahmad ali
	cout<<a;
		ahmad

6.	char s[10]; s[0]='h'; s[1]='a'; s[2]='\0'; s[3]='b'; cout<<s;	ha

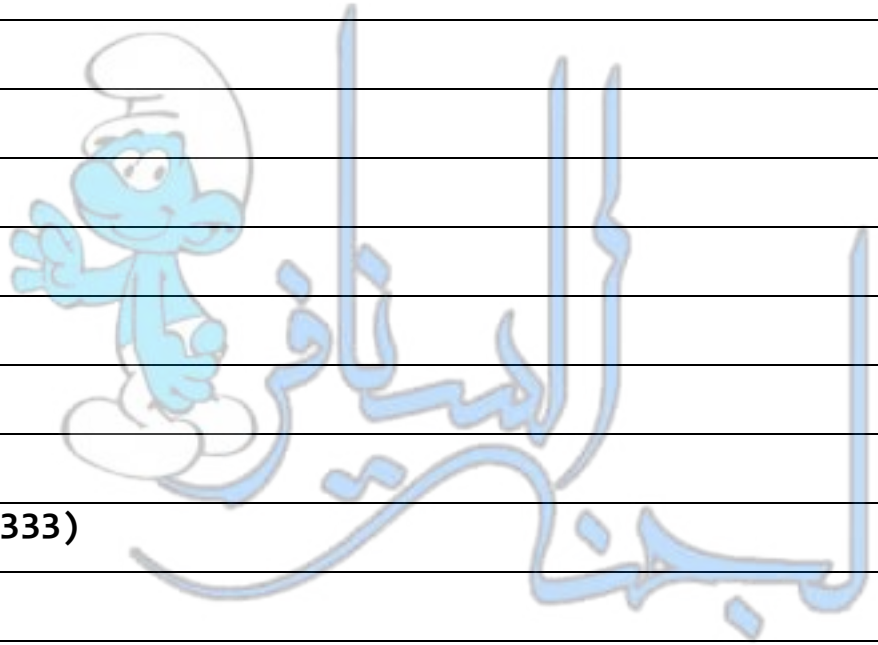
STRING FUNCTIONS		

1.	#include <iostream>
	#include <string.h>
	int main ()
	{
	char str1[10] = "Hello";
	char str2[10] = "World";
	char str3[10];
	int len ;
	strcpy(str3, str1);
	cout << "strcpy(str3, str1) : " << str3 << endl;
	strcat(str1, str2);
	cout << "strcat(str1, str2): " << str1 << endl;
	len = strlen(str1);
	cout << "strlen(str1) : " << len << endl; }
		strcpy(str3, str1) :
		Hello
		strcat(str1, str2):
		HelloWorld
		strlen(str1) :10

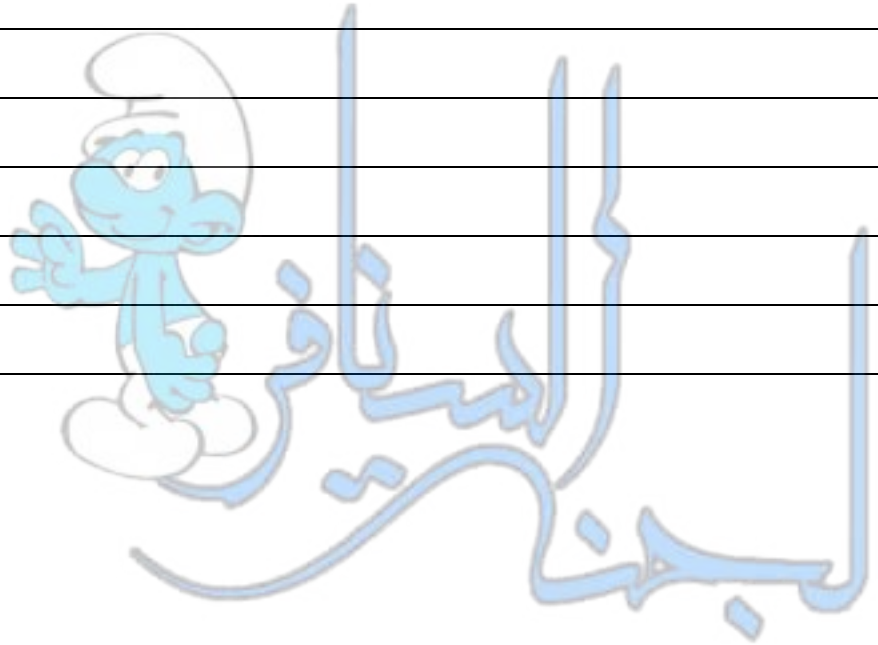
2.	#include <iostream> #include <string.h> int main () { char str1[10] = "ahmad"; char str2[10] = "Ahmad"; char str3[10] = "ali"; char str4[10] = "ali"; cout<<strcmp(str1,str2)<<'\t'; cout<<strcmp(str1,str3) <<'\t'; cout<<strcmp(str3,str4);}	1 -1 0



MATH FUNCTIONS		
1.	fabs(5.1)	5.1
2.	fabs(-5.1)	5.1
3.	abs(5.1)	5
4.	abs(-5.1)	5
5.	ceil(9.2)	10
6.	ceil(-9.8)	-9
7.	floor(9.8)	9
8.	floor(-9.8)	-10
9.	sqrt(900)	30
10.	sqrt(9)	3
11.	fmod(13.657,2.333)	1.992
12.	pow(2,-1)	0.5
13.	sin(0)	0
14.	tan(0)	0



15.	cos(0)	1
16.	exp(1)	2.71828
17.	exp(2)	7.38906
18.	log(2.71828)	1
19.	log(7.38906)	2
20.	log10(10)	1
21.	log10(1)	0
22.	log10(0.01)	-2
23.	pow(2,7)	128
24.	pow(9,0.05)	3



USER DEFINED FUNCTIONS		

1.	#include <iostream>
	void f2(int);
	void f1(){
	cout<<1;
	f2(1);
	cout<<2;}
	void f2(int x)
	{cout<<x+2;}
	void main()
	{
	f1();
	getch();}
		132



2.	//*global & local
	#include <iostream>
	int test(int,int,int);
	int a=2;int b=3;
	void main()
	{int b=2;int c=4;int d;
	d=test(a,b,c);
	b=b+3;
	cout<<a<<'\t'<<b<<'\t'<<c<<'\t'<<d;
	getch();}
	int test(int z,int x,int y)
	{z=2;
	x=b;
	a=y+a;
	cout<<endl<<z<<'\t'<<--x<<'\t'<<y++<<'\t'<<a<<endl;
	return y-x;}
		2 2 4 6
		6 5 4 3

3.	//global & local
	#include <iostream>
	int fun(int j)
	{
	j=17;
	return j;
	}
	void main()
	{int j=99;
	cout<<fun(j)<<endl;
	cout<<j;
	getch();}
		17
		99

4.	//Global & local
	#include <iostream>
	int E=22;
	void fun(int);
	void main()
	{int x=10;
	fun(x--);
	fun(x--);
	getch();}
	void fun(int v)
	{cout<<++E<<' '<<v<<' ';}
		23 10 24 9

5.	**//Call by reference and call by value** **#include <iostream>** **int fun(int &x,int y)** **{x--;** **y=y/2;** **cout<<x<<'\t'<<y<<'\t';** **return y+x;}** **void main()** **{int a=1,b=2;** **while(b>0)** **{b=fun(a,b);** **cout<<a<<'\t'<<b<<endl;** **}** **getch();** **}**	**0 1 0 1** **-1 0 -1 -1**



6.

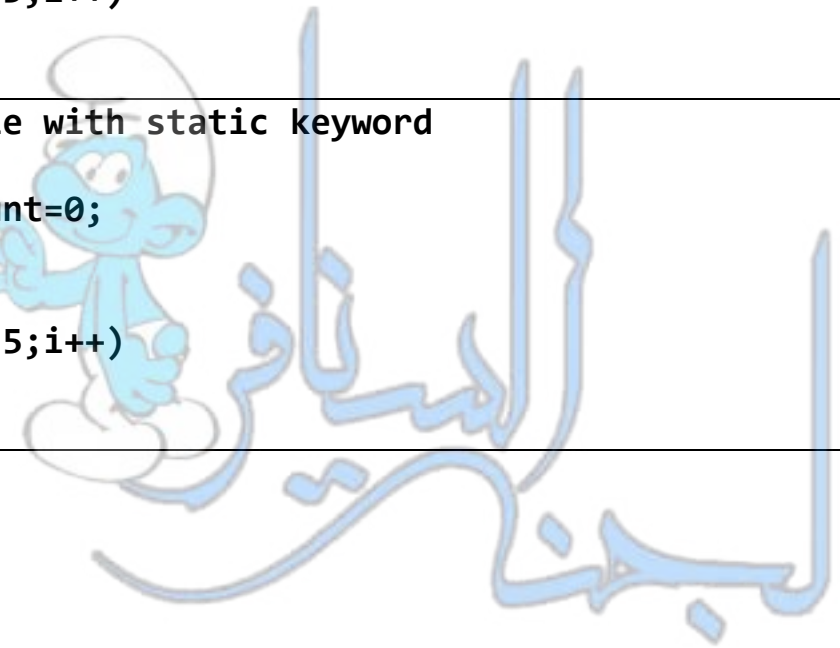
```
//Call by reference, call by value
//global & local variables
#include <iostream>
int A,B,C,D;
void two(int,int, int &);
void one (int,int, int &);
void main()
{A=6;B=4;C=3;D=4;
one(A,B,C);
cout<<A<<'\t'<<B<<'\t'<<C<<'\t'<<D<<endl;
two(D,C,A);
cout<<A<<'\t'<<B<<'\t'<<C<<'\t'<<D<<endl;
getch();}
void two (int A,int B,int &D)
{B=31;D=13;
cout<<A<<'\t'<<B<<'\t'<<C<<'\t'<<D<<endl;}
void one (int A,int B,int &C)
{int D; A=5;B=6;C=12;D=13;}
```

```
6    4    12    4
4    31   12    13
13   4    12    4
```

7.	**//Global & local variables** **#include <iostream>** **void f1(void);** **void f2(void);** **int a=6;** **int z=4;** **void main()** **{int a=2;** **{int a=7;** **cout<<a<<endl;** **cout<<z<<endl;}** **cout<<a<<endl;** **f1();** **f2();** **getch();}** **void f1()** **{cout<<a<<endl;}** **void f2()** **{int a=3;** **cout<<a<<endl;}**	**7** **4** **2** **6** **3**

8.	//counter example without static keyword
	void counter()
	{ int count=0;
	cout<<count++;}
	void main()
	{for (int i=0;i<5;i++)
	counter();
	getch();}
		00000

9.	//counter example with static keyword
	void counter()
	{ static int count=0;
	cout<<count++;}
	void main()
	{for (int i=0;i<5;i++)
	counter();
	getch();}
		01234



10.	// Function calling Array
	#include <iostream>
	int test (int a[],int size)
	{int sum=0;
	for(int i=1;i<size;i++)
	{sum+=a[i];
	a[i]=a[i]*2;}
	return sum;}
	void main()
	{int x[10]={2,1,2,1,1,2,3,2,1,2};
	cout<<x[2]<<endl;
	cout<<test(x,5)<<'\t';
	for(int i=1;i<10;i++) cout<<endl<<x[i];
	getch();
	}
		2
		5
		2
		4
		2
		2
		2
		3
		2
		1
		2

11.	#include <iostream> int fun(char ss1[], char ss2[]) {int a=1,b=3; strcpy(ss1,ss2); cout<<ss1; return b%a;} void main() {char s1[20]="comp"; char s2[20]="science";	science10science

	int x=1;
	int y;
	y=fun(s1,s2);
	cout<<x<<y<<s1;
	getch();}	

12.	//multiple cin
	#include <iostream>
	main()
	{char s1[20];
	cin>>s1;//ahmad ali
	cout<<'1'<<s1<<endl;
	cin>>s1;//yara
	cout<<'2'<<s1 <<endl;
	cin>>s1;//sarah
	cout<<'3'<<s1 <<endl;}
		ahmad ali
		(cin1+cin2)
		1ahmad
		2ali
		yara (cin3)
		3yara



REVISION		

1.	//scopes
	#include<iostream.h>
	int x=3;
	void main()
	{ int x=9;
	{
	int x=21;
	cout<<x;}
	}
		21

2.	// function type casting #include <iostream.h> int fun(double x, double y) { return x-y;} void main() {cout << fun(6.4, 2.1);}	4

3.	//call by reference & call by value
	#include<iostream.h>
	void fet(int& f, int g)
	{
	if(g<f)
	{ f++;
	g++; }
	if(f<g)
	{ f--;
	g--;} }
	void main()
	{int x=7,y=8;
	fet(x,y);
	cout<<x<<y;}
		68

4.	// string.h
	#include<iostream.h>
	#include <string.h>
	void main()
	{char s1[20]="albalqa ";
	char s2[20]="applied ";
	char s3[20]="university ";
	strcpy(s1,s2);
	strcat(s1,s3);
	cout<<s1;}
		applied university

5.	// static keyword
	#include<iostream.h>
	int fun1 (int z)
	{static int x = 3;
	x *= z ;
	return x;}
	void main ()
	{int y;
	for (int i=3;i<5;i++)
	y = fun1 (i);
	cout<<y;}
		36

6.	//USER function & math functions
	#include<iostream.h>
	#include<math.h>
	int fun1(int x)
	{ return pow(x,2);}
	int fun2(int x)
	{ return sqrt(x);}
	void main()
	{ int x=3;
	int y;
	y=fun1(x);
	cout<<fun2(y);}
		3

7.	// strings & string functions #include <iostream.h>	MNOP

```
#include<string.h>
void array(char[]);
void main()
{    char arr[] = {'L', 'M', 'N', 'O', '\0'};
     array(arr);
     cout << arr;}
void array(char x[])
{    for(int i=0; i < strlen(x); i++)
          x[i] +=1;}
```



8.	// functions & do while
	#include<iostream.h>
	int fun(int x)
	{
	if(x==0 ||x==1)
	return x;
	else
	return (x-1)*4;
	}
	void main()
	{
	int i=2;
	do
	{
	cout<<fun(i)<<" ";
	i++;
	}
	while(i<=4);
	}
		4 8 12



9.	// array & function
	#include<iostream.h>
	const int size=4;
	void mult(int []);
	void main()
	{ int a[size]={3,4,5,6};
	mult(a);
	cout<<a[0];}
	void mult(int x[])
	{for(int i=0;i<size;i++)
	x[i]*=2;
	cout<<x[0];}
		66

10.	// nested if-else void main() { int x=1; if(!x) cout<<"*"; else if(1-x) cout<<"&"; else if(x-1) cout<<"@"; else cout<<"+";}	+

11.	// if
	int b=1;
	if(b=5)
	cout<<b<<endl;
	else
	cout<<b-1<<endl;
		5

12.	// while & math functions
	#include<iostream.h>
	#include <math.h>
	void main()
	{
	int j=1;
	float i=1.5;
	bool x = 1;
	while(x)
	
	{ cout << floor(i) << " ";
	i += 1.4;
	x *=j;
	j--; }
	cout << ceil(i) << " ";}
		1325

13.	**// string functions** **#include<iostream.h>** **#include <string.h>** **void main()** **{char str1[20]="Final Exam";** **char str2[20]="Final exam";** **int x= strcmp(str1, str2);** **if (x == 0)** **cout<<"&&&";** **else if (x == 1)** **cout<<"!!!";** **else** **cout<<"@@@";** **cout<<strlen(str1);}**	**@@@10**
14.	**// function calling function** **#include<iostream.h>** **float f1(int x)** **{ return x++;}** **char f2(float y)** **{ return y+2.5;}** **float f3(int n, int m)** **{ cout<<f1(n) <<f2(m)<< '\t';** **return f1(n) + f2(m);}** **void main() { cout<<f3(103,'d')/2;}**	**103f 102.5**

15.	// nested for
	char ar[5] = {'\0'};
	int k = 3;
	int matx[2][2] = {67, 68, 69, 70};
	for(int i=0; i<2; ++i)
	for(int j=0; j<2; ++j)
	ar[k--] = matx[j][i];
	cout << ar;
		FDECP

16.	//2D ARRAY int y[3][3] = {1, 2, 3, 4, 5, 6, 7, 8, 9}; cout << y[1][2];	6

17.	//1D ARRAY
	int list[5];
	for (int i = 0; i < 5; i++)
	{list[i] = i * i + 5;
	if (i % 2 == 0)
	list[i] = list[i] + i;
	else
	list[i] = list[i] - i;
	cout <<list[i]<<" ";}
		5 5 11 11 25

18.	// while
	int x=7, y=3;
	while(y!=x)
	{x = x-1;
	y= y+1 ;}
	cout<<x<<y;
		55

19.	//2D ARRAY void main() { char m[4][4]={{'a','b','c','d'},{'e','f','g','h'},{'i','j','k','l'}, {'m','n','o','p'}}; for(int i=0;i<4;i++) cout<<m[i][3]<<" ";}	d h l p

20.	// SWITCH
	void main()
	{
	int x = 3,y;
	switch(x+3)
	{ case 6: y=2;
	case 7: y*=3;
	default: y+=1;}
	cout<<y;
	}
		7



21.	// continue
	void main()
	{
	float x=10;
	for(int j=0;j<3;j++)
	{
	x/=2;
	if(x>2)
	continue;
	cout<<"c++"<<" ";}}
		C++

22.	// break #include<iostream.h> void main() { for(int i =0;i<2;i++) for(int j=0;j<2;j++) { if(j==1) break; cout<<j<<" ";}}	0 0
23.	//1D ARRAY void main() { int std[3]={10,14,25}; for(int i=0;i<2;i++) std[i]=std[i]-std[i+1]; cout<<std[0]<<" "<<std[1]<<" "<<std[2];}	-4 -11 -25

24.	//1D ARRAY int trans[]={30,31,32}; for(int i=0; i<=2; i++) {trans[i]=trans[2-i]; cout<<trans[i]<<" "; }	32 31 32
25.	//2D ARRAY if the array is declared as follows: int a[3][4]={2,5,8,1,4,7,3,6,9,10,11,12}; the second element in the third row is?	10
26.	// strings char a[]={'E','N','G','I','N','E','\0','R','I','N','G'}; cout<<a;	ENGINE

27.	//1D-ARRAY & BOOL
	void main( )
	{
	bool a[3]={0,44};
	cout<<a[0]<<" "<<a[1];
	}
		0 1

28.	//1D ARRAY void main( ) { bool a[3]={7,0}; cout<<a[0]<<" "<<a[1];}	1 0

29.	**//1D ARRAY** **void main( )** **{** **int a[3]={7.8,9.3};** **cout<<a[0]<<" "<<a[1];** **}**	**7 9**
30.	**//1D ARRAY** **void main( )** **{ int a[3]={1.2,3.5};** **cout<<a[0]<<" "<<a[1];}**	**1 3**
31.	**//scopes** **#include<iostream.h>** **int a=4, b=2;** **void main()** **{ int a,c=6;** **a=b++;** **{ int a=13,z;** **z=a + ++b + c--;** **cout<<z<<endl; }** **cout<<a<<b<<c;}**	**16** **13-24**

32.	**//do-while** **int x = 1;** **do** **{** **if (x%3 != 0)** **x += 2;** **else x--;** **cout << x << "+";** **} while (x < 7 && x >= 2);**	**3+2+4+6+5+7+**
33.	**////1D ARRAY** **int a[4]; a[0] = 2;** **for (int i = 1; i < 4; i++) a[i] = i + a[i-1];** **for (i = 0; i < 4; i++) cout << a[i] << " ";**	**2 3 5 8**
34.	**//for loop** **int i = 0, j = 12;** **for (; i < j; i += 3, j -= 4)** **cout<< i * j;** **cout<< i<< j;**	**02464**

GOOD LUCK